

Air Quality & Health Impact Analysis

MACHINE LEARNING APPROACH

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PROJECT OBJECTIVE

Polluted air can lead to increased risk of various diseases such as asthma, chronic heart diseases and more. Thus, clean air has now become a necessity for good health and life.

Classify health impacts using air quality data with machine learning

LITERATURE REVIEW

Gupta et al. (2023): Random Forest and SVM for AQI prediction

Compared various machine learning models, including Random Forest and Support Vector Machines (SVM), to predict AQI. Their study emphasizes how machine learning can significantly improve the accuracy of air quality forecasts, a critical factor for informing public health and environmental policies.

Gore & Deshpande (2017): Decision Trees and Neural Networks.

Demonstrated the successful application of Decision Trees and Neural Networks for forecasting air quality levels. Their findings underline the importance of selecting the right model and configuration to achieve reliable predictions, particularly for real-time air quality monitoring and public health assessments

Liang et al. (2020): Gradient Boosting and k-NN.

Applied Gradient Boosting and K-Nearest Neighbors to forecast air quality levels. Their study revealed how machine learning models could not only predict air quality but also highlight the most significant pollutants contributing to poor air quality. This research provides valuable insights into the use of machine learning for air quality management and health risk mitigation.

There is a limited body of literature that integrates air quality data with health impact predictions. While a large volume of research has focused on air quality prediction alone, fewer studies have examined its direct impact on health outcomes

DATASET DESCRIPTION

Kaggle's Air Quality and Health Impact Dataset

This dataset contains comprehensive information on the air quality and its impact on public health for 5,811 records.

Features:

- Air Quality Metrics: AQI, PM2.5, NO2, etc.
- Weather Conditions: Temperature, Humidity, etc.
- Health Metrics: Respiratory and Cardiovascular Cases.

Target: HealthImpactClass



HEALTH IMPACT CLASS

80

Classification of the health impact based on the health impact score:

60

0: Very High (≥ 80)

1: High ($60 < 80$)

2: Moderate ($40 < 60$)

3: Low ($20 < 40$)

4: very Low (< 20)

40

20

M E T H O D O L O G Y

- **Preprocessing:**

- Min-Max normalization for scaling features
- PCA for dimensionality reduction (retaining 95% variance)

- **Models:**

- Artificial Neural Networks (ANN)
- Support Vector Machines (SVM)
- Decision Trees (DT)
- k-Nearest Neighbors (k-NN)

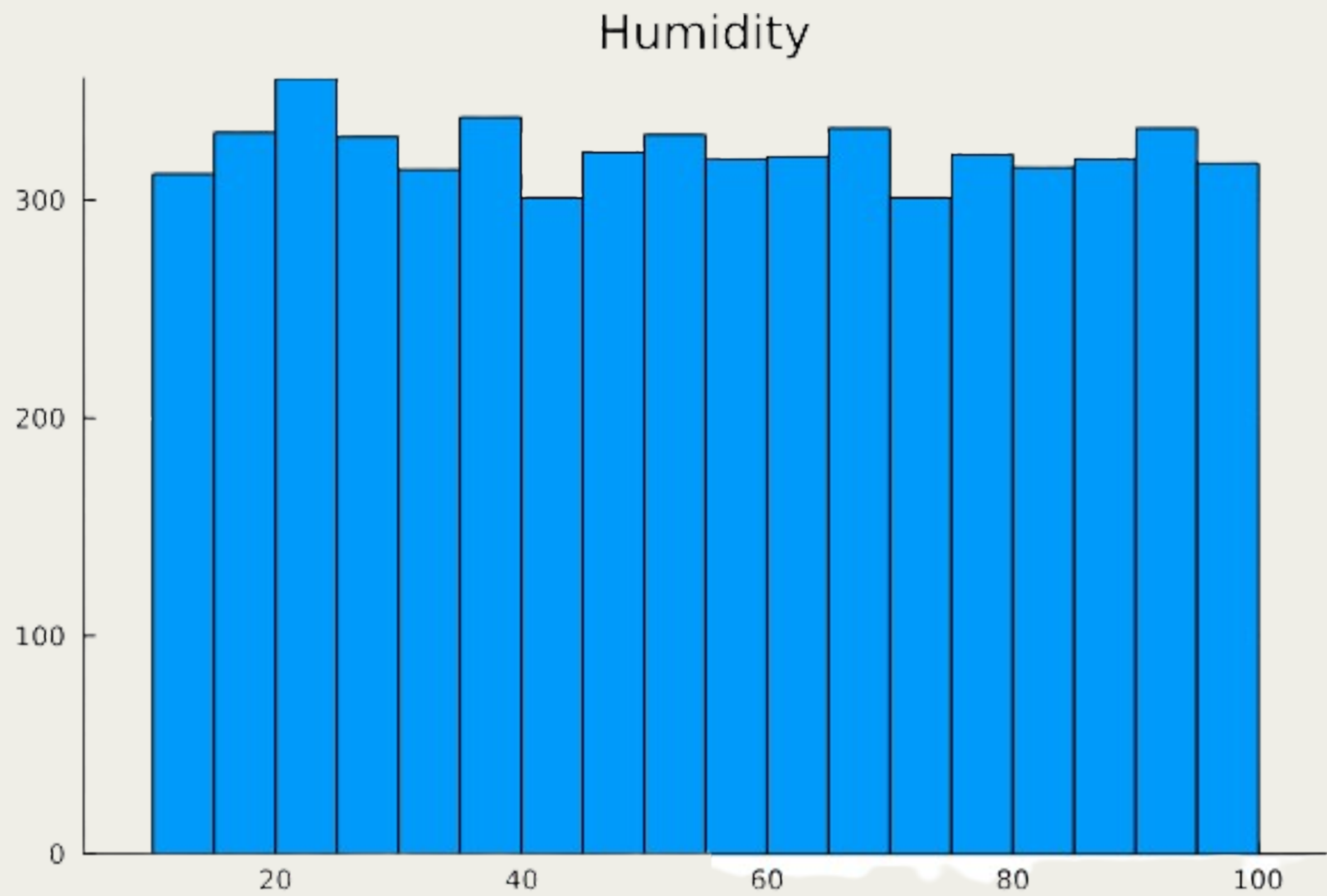
- **Ensemble Method:**

- Stacking classifier for improved accuracy

OUR APPROACHES

Approach	Accuracy	Error Rate	F1-score
MinMax	94.23%	5.76%	2.856
ZeroMean	93.29%	6.70%	2.483
CrossVal	94.40%	5.59%	2.860
PCA_3	82.74%	17.25%	0.905
PCA_095	93.67%	6.32%	2.594

FEATURES DISTRIBUTION



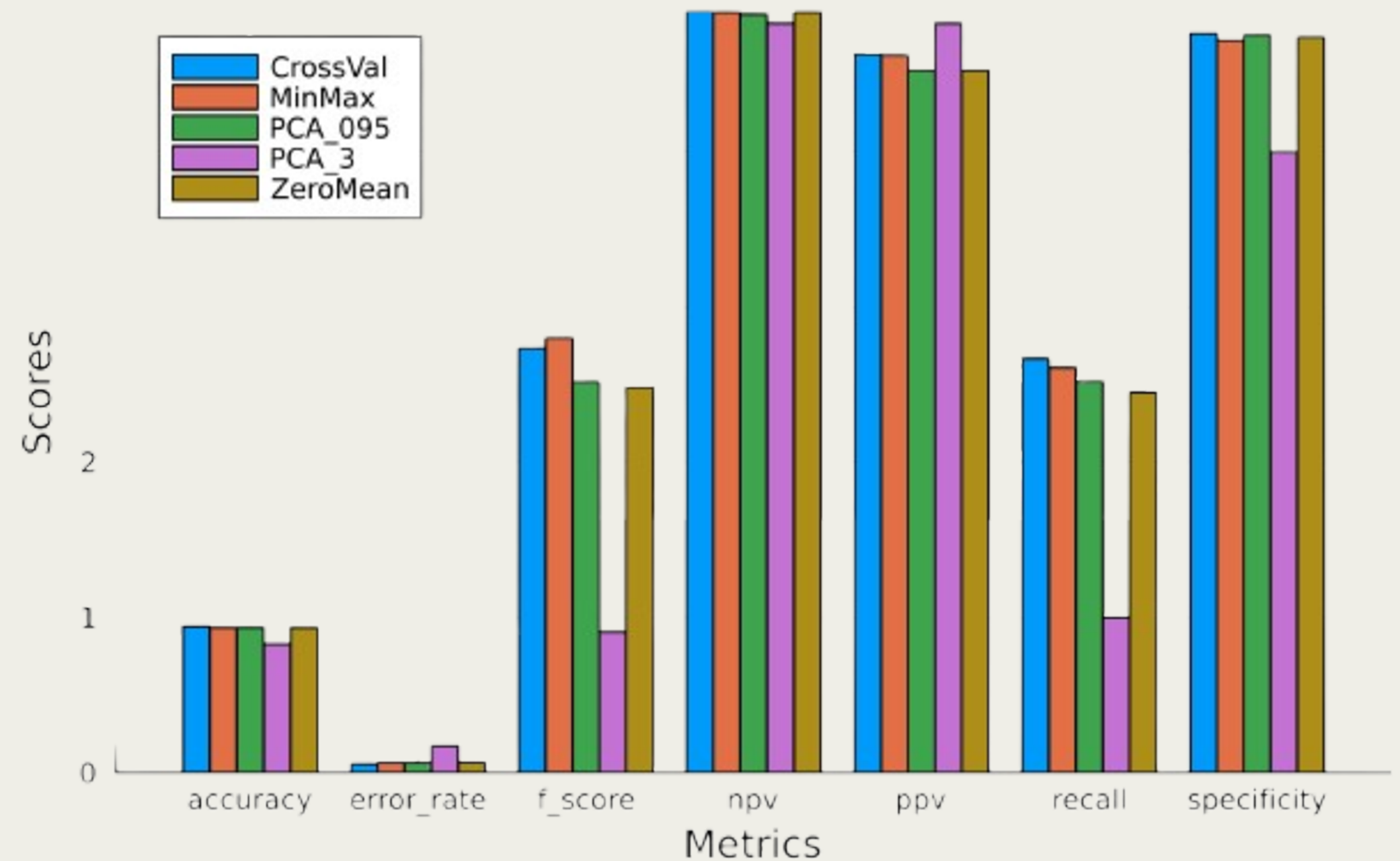
BEST PERFORMANCE

Cross-validation with Stacking

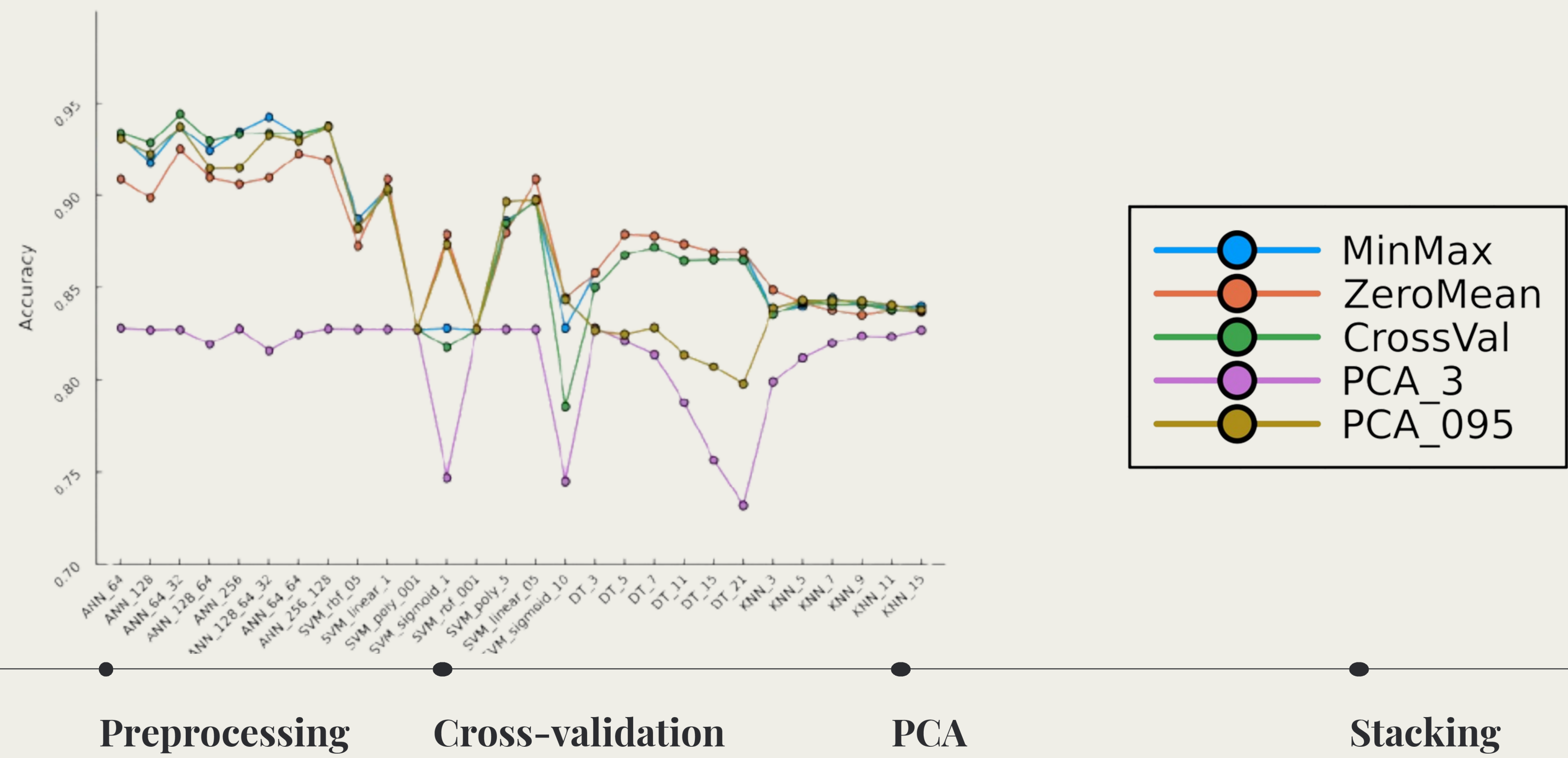
94.40%
Accuracy

5.59%
Error Rate

2.86
F1-Score



DISCUSSION



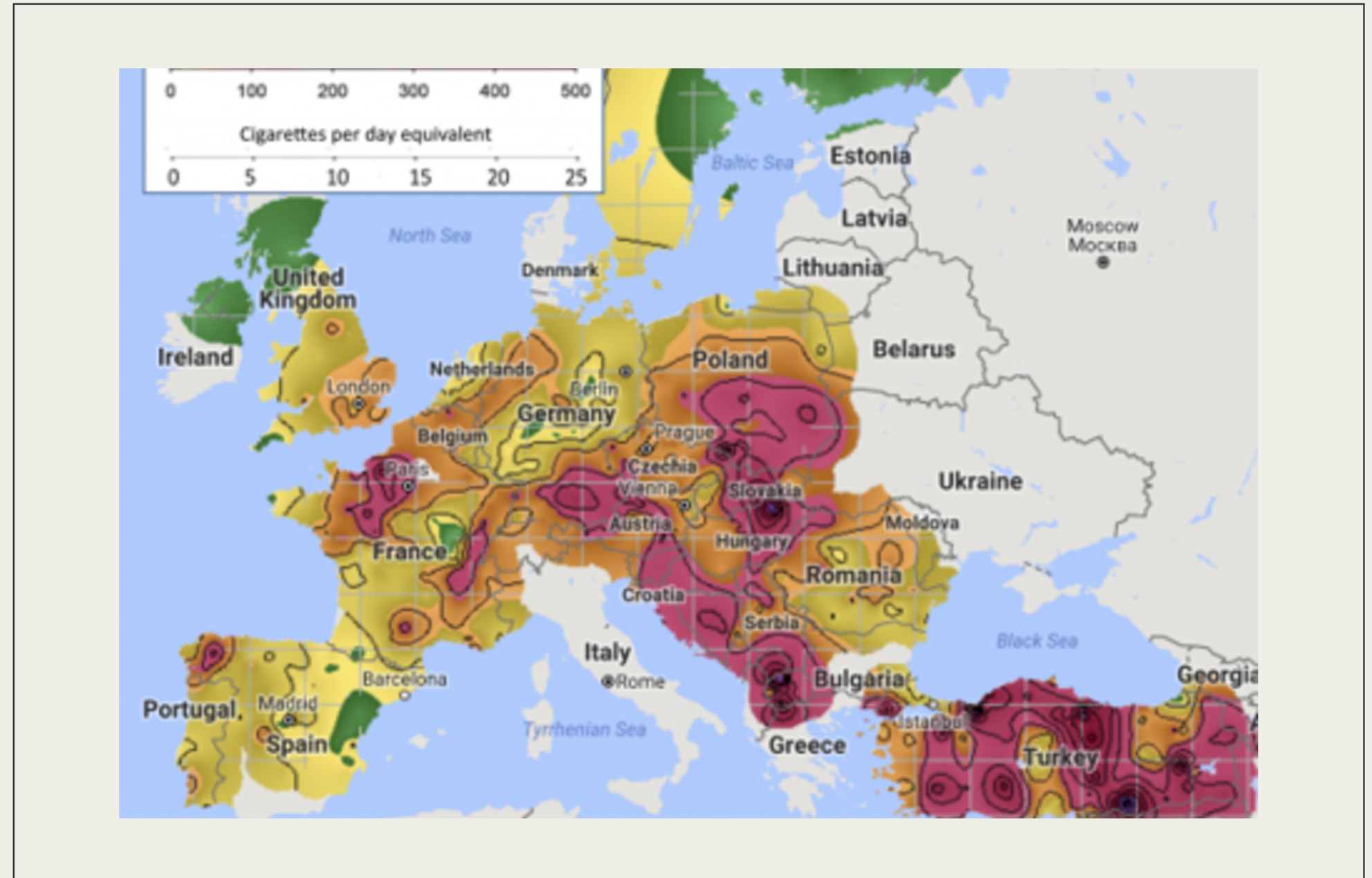
CONCLUSION

Key Findings:

- Cross-validation with stacking: Best performance.
- Min-Max normalization: Effective for scaling.
- PCA with 95% variance: Balanced dimensionality reduction.

Future Directions:

- Use real-time air quality data.
- Include additional environmental and health factors.
- Optimize ensemble methods for scalability.



Thank you!

**SPECIAL THANKS TO OUR TEAM WHO TOOK TURNS SLEEPING A
COUPLE OF HOURS A DAY**

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